



Department of Mathematics, Physics, and Computer Science
College of Science and Mathematics

UNIVERSITY OF THE PHILIPPINES MINDANAO

Rm 220, CSM Bldg., UP Mindanao, Mintal, Davao City 8022
T: + 63 82 2930303 E: dmpcs-csm.upmindanao@up.edu.ph

Degree Program BS Computer Science
Date Effective 1 June 2018
Approved by UP Board of Regent

COURSE SYLLABUS IN CMSC 18 – COMPUTER PROGRAMING I

UP Statement of the Philosophy of Education and Graduate Attributes <i>Approved by the Board of Regents on 28 November 2019</i> A UP education seeks to produce graduates imbued with an abiding sense of responsibility to their people and nation, the skills and mindsets to improve human life, and a commitment to the freedom and welfare of all. Aside from mastery of knowledge in their specific disciplines, UP graduates must possess breadth of mind, strength of character, and generosity of spirit, fostered by a firm grounding in both the arts and sciences, and such specialist courses as their programs may require. They must be prepared to inclusively engage with society and the world at large, mindful of their people’s needs and capabilities, and keen to the challenges and opportunities of national development in this century of rapid global change. UP aims to achieve this through its General Education program, one that develops mind, body and spirit, which familiarize all its students with their culture and history and fosters a sense of shared citizenship, while equipping them with critical thinking, discernment and technical skills they will need to excel in their chosen professions.	UP Mindanao Vision An internationally competitive graduate, research, and public service university in Mindanao, leading in the study of arts, cultures, and sustainable biosystems. Mission The University of the Philippines Mindanao provides distinctive interdisciplinary education, research & development, and public engagement that are responsive and relevant to the diverse sectors in Mindanao and sustains Mindanao heritage and biodiversity through innovative systems and technologies. Department Vision The department’s vision is to become the Center of Excellence for Mindanao’s development through Mathematics and Computing Sciences. Department Mission The department’s mission is to excel in Mathematics and Computing Sciences by a) providing world class quality education, impactful research, and responsive extension services to diverse sectors in Mindanao and the rest of Southeast Asia through its academic programs in the sciences and engineering, b) cultivating passion for excellence, creative thinking, and commitment to the common good in the context of culturally diverse global community.	Program Learning Outcomes <table><tr><td>PLO 1</td><td>Apply appropriate knowledge, skill and techniques in computer science for real-world problem-solving, and conducting multi-disciplinary research.</td></tr><tr><td>PLO 2</td><td>Design and develop IT systems that align with industry demands by integrating computer science knowledge and skills with interdisciplinary expertise.</td></tr><tr><td>PLO 3</td><td>Communicate effectively in oral and written communication to both technical and non-technical audiences.</td></tr><tr><td>PLO 4</td><td>Demonstrate adaptive lifelong learning skills in acquiring new knowledge, skills, and techniques in computer science.</td></tr><tr><td>PLO 5</td><td>Demonstrate leadership, professionalism, and the capability to collaborate in a diverse multi-disciplinary team.</td></tr></table>	PLO 1	Apply appropriate knowledge, skill and techniques in computer science for real-world problem-solving, and conducting multi-disciplinary research.	PLO 2	Design and develop IT systems that align with industry demands by integrating computer science knowledge and skills with interdisciplinary expertise.	PLO 3	Communicate effectively in oral and written communication to both technical and non-technical audiences.	PLO 4	Demonstrate adaptive lifelong learning skills in acquiring new knowledge, skills, and techniques in computer science.	PLO 5	Demonstrate leadership, professionalism, and the capability to collaborate in a diverse multi-disciplinary team.
PLO 1	Apply appropriate knowledge, skill and techniques in computer science for real-world problem-solving, and conducting multi-disciplinary research.											
PLO 2	Design and develop IT systems that align with industry demands by integrating computer science knowledge and skills with interdisciplinary expertise.											
PLO 3	Communicate effectively in oral and written communication to both technical and non-technical audiences.											
PLO 4	Demonstrate adaptive lifelong learning skills in acquiring new knowledge, skills, and techniques in computer science.											
PLO 5	Demonstrate leadership, professionalism, and the capability to collaborate in a diverse multi-disciplinary team.											

COURSE INFORMATION

Course Number:	CMSC 18
Course Title:	Computer Programming I
Course Description:	Elements in Programming – Design and Analysis of Algorithms, Basic Concepts in Programs Development (Control Structures, Arrays, Structures, Functions, Text Files, Recursion, Pointers, Data Structures)
Course Prerequisite:	NONE
Course Credit:	3 units
Offering period	First Semester

COURSE LEARNING OUTCOMES

At the end of the course, the students should be able to formulate solutions for a variety of problems through implementation of well-structured programs in the C/C++ Language. Specifically, the students should be able to:

<i>Program Learning Outcomes aligned to the CLOs</i>		
CLO 1	Explain the formulated solutions in form of algorithms;	PLO 1,2
CLO 2	Demonstrate their coding capabilities from their understanding on the basic concepts in program development such as declaration, input and printing, arithmetic, functions, and files; and	PLO 3,4
CLO 3	Demonstrate their implementation capabilities in the C/C++ language through proper coding of simple and structured data types program control structures, arrays, loops, recursion, text files, structures, functions, pointers, and basic data structures based from their formulated solutions.	PLO 5

COURSE OUTLINE

1. Elements of Programming

1.1. Design of Algorithms

- 1.1.1. Problem Solving with a Computer
- 1.1.2. Introduction to Algorithm Development
- 1.1.3. Representation of Algorithms
 - 1.1.3.1. Pseudocode
 - 1.1.3.2. Flowchart

1.2. Implementation of Algorithms (Programming)

1.2.1. Representation of Data

- 1.2.1.1. Simple Data Types
- 1.2.1.2. Structured Data Types
- 1.2.1.3. Variable Declaration Arrays
- 1.2.1.4. Characters and Strings Basics

1.2.2. Statements, Expressions, and Operators

1.2.3. Basic Program Control Structures

- 1.2.3.1. Sequence
- 1.2.3.2. Selection (if, else, else if)
- 1.2.3.3. Repetition (for, while, do-while)
- 1.2.3.4. Selection (switch)
- 1.2.3.5. Loops
 - 1.2.3.5.1. Counter Controlled Loops
 - 1.2.3.5.2. Event Controlled Loops
 - 1.2.3.5.3. Infinite Loops
 - 1.2.3.5.4. Continue and goto

1.2.4. Arrays

- 1.2.4.1. Single Dimension Arrays
- 1.2.4.2. Multi Dimension Arrays

- 1.2.4.3. Initialization and Size of Arrays
- 1.2.4.4. Arrays as Pointers
- 1.2.4.5. Pointer Operations
- 1.2.4.6. More on char and strings

1.2.5. Structures

- 1.2.5.1. Initialization using Structures
- 1.2.5.2. Applications of Structures

1.2.6. Functions and Structured Programming

- 1.2.6.1. Function Declaration and Pass by Value
- 1.2.6.2. Pass by Reference
- 1.2.6.3. Recursion

1.2.7. File Manipulation

MODE OF DELIVERY

For Lecture Part - The class meets twice a week only which are spread over 16-week period week period (32 lecture sessions). Each session will have a time duration of 1 hour.

For Laboratory Part - The class meets once a week for the 16-week period (16 laboratory sessions). Each session will have a time duration of 3 hours

COURSE MATERIALS

To facilitate the learning of this course, students need the following:

Material Name	Description
Course Guide	This document contains the course description, learning outcomes, study schedule, course requirements, and grading scheme. This must be read at the start of classes and, when necessary, revisited to keep the student on track with the progression of the course.
Study Guide	This is provided for each segment of topic, which may cover a week or two, to guide the students in their weekly activities and learning progress.
Lecture Presentations	Lecture slides will be given in PDF format (you may need to install a PDF reader for this). This will supplement the video lectures.
Video Lectures/ Tutorials	These are lectures or tutorials of the topics covered. Students need to watch these videos, which are created by your professor or educators from other universities or experts from the industry. Links are provided in the Study Guide posted in the LMS.
Computer	Any device e.g. desktop, laptop, tablet or smartphone, which has connection to the Internet to access the learning materials.
Integrated Development Environment (IDE)	Any IDE of their preferred programming language for the development of their image and video processing software project.

LEARNING ACTIVITIES

Study schedule in **Lecture and Laboratory**

Week #	Session #	Course Outcomes	Topics	Activities	References	Assessment
Week 1-2	Session 0	Acquire an overview of the course syllabus and class policy	Course Syllabus	Open discussion	Course Syllabus	Oral recitation
	Lecture Session 1	- solutions to given problems into algorithms - differentiate pseudocodes from flowcharts - design pseudocodes and flowcharts based on their algorithms	Introduction to Algorithm Development	Lecture Guided Open discussion	[1],[2],[3],[4],[5],[6],[7],[8]	Oral recitation
	Lecture Session 2		Pseudocode	Guided Open discussion w/ Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Written exercise Take-home Exercise
	Lecture Session 3		Flowchart	Guided Open discussion w/ Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Written exercise Take-home Exercise
	Laboratory Session 1		Pseudocode and Flowchart	Laboratory Exercise on creating Flowcharts and Pseudocodes	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Written Exercises
Week 3	Lecture Session 4	- describe different basic variable types and constants in programming - differentiate the legal from the illegal variable declarations in C/C++ - demonstrate the declarations of variable types in C/C++ code - construct a fully functional code in C/C++ using discussed concepts	- Basic Computer Memory - Variables and Keywords in C/C++ - Numeric Variable Types - Variable Declarations	Lecture Program Tracing Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation
	Lecture Session 5		- Constants - String Basics – Reading Strings from keyboard	Lecture Program Tracing	[1],[2],[3],[4],[5],[6],[7],[8]	Written exercise Take-home Exercise
	Laboratory Session 2		Variable Types and Constants	Laboratory Exercise Variable Types and Constants (programming)	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises

Week 4	Lecture Session 6	- Distinguish Statements from Expressions - Convert Expressions into proper C/C++ Statements - Demonstrate the correct use of the different operators and their precedence in Statements based on the algorithm at hand	- Statements and White Spaces - Null and Compound Statements - Expressions - Assignment Operators - Unary and Binary Mathematical - Basic Operators and Precedence	Lecture Guided Open discussion with Sample Basic Problems Program Tracing	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation
	Lecture Session 7		- Relational and Logical Operators and Precedence - Compound Assignment Operators - Conditional Operators and Precedence	Lecture Guided Open discussion with Sample Basic Problems Program Tracing	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation Take-home Exercise
	Laboratory Session 3		Statements, Expressions, and Operators	Laboratory Exercise on Statement Declarations and Operators	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 5	Lecture Session 8-9	- Demonstrate the correct implementation for program control statements - Identify when to use if, if-else, else-if in designing algorithms for applicable problems - Identify when to use for loops, do-while, and while in designing algorithms for applicable problems - Identify which among the program control statements provide the most optimal programming solution to answer given problems	- Program Control Statements - Sequential Control - Selection Control (single, two, three, multiple alternatives) - Dangling Else Statements	Lecture Guided Open discussion with Sample Basic Problems Program Tracing	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation Written Exercise
	Laboratory Session 4		Program Control Statements (Sequential and Selection Control)	Laboratory Exercise on Program Control Statements 1	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 6	Lecture Session 10-11		Repetition Control (for loops, do-while, while)	Guided Open discussion with Sample Basic Problems Program Tracing	[1],[2],[3],[4],[5],[6],[7],[8]	Take-home Exercise
	Laboratory Session 5		Program Control Statements (Repetition Control)	Laboratory Exercise on Program Control Statements 2	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises

Week 7	Lecture Session 12	Demonstrate understanding of the concepts on Basic Programming Elements (Variable declarations, Statement Declarations, Operators, Program Control Statements)	1 st Long Exam	Review for LE1	[1],[2],[3],[4],[5],[6],[7],[8]	
	Lecture Session 13			Written Exam		Comprehensive Exam
	Laboratory Session 6		1 st Laboratory Exam	Programming Exam		Comprehensive Programming Exam
Week 8-9	Lecture Session 14-15	- Demonstrate the correct implementation of Arrays (Naming, Declaring, Initialization and Manipulation) - Distinguish when to use arrays and pointers in implementing algorithms to solve given problems - Explain the different ways to access the array - Demonstrate the correct implementation of Pointers in Array use	- Naming and Declaring Arrays - Single and Multidimensional Arrays - Initialization of Arrays - Size of Arrays	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation Written Exercise
	Laboratory Session 7		Arrays (basics)	Laboratory Exercise on Arrays 1	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Written Exercise
	Lecture Session 16-17		- Arrays as Pointers - Pointer Manipulations	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation Take home exercise
	Laboratory Session 8		Arrays (more)	Laboratory Exercise on Arrays 2	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 10	Lecture Session 18-19	- Differentiate the implementation of char and strings in C/C++ codes - Explain and implement the ways to access characters within strings - Demonstrate the correct implementation of Pointers in string manipulations - Identify the different techniques in string manipulations	More on Characters and Strings	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Written exercise
	Laboratory Session 9		The char Data Type ASCII Table Manipulating strings			
			Characters and Strings	Laboratory Exercises on Characters and Strings	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 10	Lecture Session 20	Demonstrate understanding of the concepts on the Programming Elements of Arrays, Characters, and Strings (one and multi-dimensional arrays, arrays as pointers, array and string manipulation)	2 nd Long Exam	Review for LE2	[1],[2],[3],[4],[5],[6],[7],[8]	
	Lecture Session 21			Written Exam		Comprehensive Exam
	Laboratory Session 10		2 nd Laboratory Exam	Programming Exam		Comprehensive Programming Exam

Week 11	Lecture Session 22-23	- Identify the structure definition itself and its members - Demonstrate the correct implementation of Structure Declarations, Instances, Manipulations, and Complex Structures. - Explain when structures can be more efficient than arrays - Exhibit structure manipulation	- Structures - Basic Structure Definition - Writing Structures and Structure Instances - Manipulating Structure Members - Complex Structures - Structures inside Structures	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitation Take home exercise
	Laboratory Session 11		Structures	Laboratory Exercises on Structures	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercise
Week 12-13	Lecture Sessions 24-27	- Properly Demonstrate the basic implementation of a program into a collection of functions - Explain the importance of implementing functions - Identify the parts of a given program that can be implemented as functions - Differentiate Parameters and Arguments and when to implement either - Define and Implement Recursion using functions	- Structured Programming and Functions - Program flow with functions - Function Calls and Events - Writing a Function - Parameters and Arguments - Passing and Returning a Value - Recursion	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitations Written exercises Take home exercises
	Laboratory Session 12-13		Structured Programming and Functions	Laboratory Exercises on Functions	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 14-15	Lecture Session 28-29	- Demonstrate proper file handling in C/C++ programming - Identify the different file handling functions - Demonstrate the correct implementation of file handling functions to open/close, input/output, and read/write to files	- Introduction to Files - File Handling Functions - File Manipulation (Opening/Closing a File, Input/Output operations on files, Reading/Writing to files.	Lecture, Program Tracing, Guided Open discussion with Sample Basic Problems	[1],[2],[3],[4],[5],[6],[7],[8]	Oral Recitations
	Laboratory Session 14	- Differentiate the different file manipulation modes and explain when to use them - Demonstrate understanding of the concepts on Structures, Functions, and File Handling	Differences in File Manipulation Modes	Laboratory Exercises on Files	[1],[2],[3],[4],[5],[6],[7],[8] Lecture Notes	Programming Exercises
Week 16	Lecture Session 31	Demonstrate understanding of the concepts on the Programming Elements of Structures, Functions, and File Handling (Declaration and initialization of structures, structure member manipulations. Complex structures, structured programming, implementing functions, function calls and events, parameters and arguments, recursion, file handling functions, and file manipulation)	3 rd Long Exam	Review for LE2	[1],[2],[3],[4],[5],[6],[7],[8]	
	Lecture Session 32			Written Exam		Comprehensive Exam
	Laboratory Session 16		3 rd Laboratory Exam	Programming Exam		Comprehensive Programming Exam

Week X	Set by OUR	- Demonstrate their capability to understand code based on the basic concepts in program development such as declaration, input and printing, simple and structured data types, arithmetic, program control structures, arrays, loops, recursion, text files, structures, functions, pointers, and all data structures - Express solutions for a variety of problems into algorithms and write these as well-structured programs readable by the C/C++ Language	Final Exam	Group project	[1],[2],[3],[4],[5],[6],[7],[8]	Programming Project Presentation
---------------	-------------------	--	------------	---------------	---------------------------------	----------------------------------

REFERENCES

- [1] Roberts, Eric S. (1995). *The art and science of C: a library-based introduction to computer science*. Addison-Wesley.
- [2] Mata-Toledo, Ramon. (2000). *Introduction to computer science*. McGraw-Hill.
- [3] Eggen, Roger and Maurice Eggen. 1994. *An introduction to computer science using C*. PWS Pub. Co.
- [4] Fuller Floyd. (2003). *Computers: understanding technology*. EMC Paradigm.
- [5] Goldschlager, L. and A. Lister. (1988). *Computer Science: A Modern Introduction (2nd ed.)*. Prentice-Hall.
- [6] Brookshear, J. Glenn. (1994). *Computer science: an overview*. Benjamin/Cummings Pub. Co.
- [7] Beekman, George. (1997). *Computer confluence: exploring tomorrow's technology*. Benjamin/Cummings Pub. Co.
- [8] Johnson, Deborah G. and Helen Nissenbaum (eds.). (1995). *Computers, ethics & social values*. Prentice Hall.

COURSE REQUIREMENTS

Lecture

Long Exams	30%
Assignment/Quizzes	10%
Final Exam	10%
	50%

Laboratory

Exercises	25%
Laboratory Exams	
(On the-Spot Programming)	25%
	50%

Total : 100%

GRADING SCALE

Final grades will be based on the following grading scheme:

[96,100]	1.00	[75, 80]	2.25
[92, 96]	1.25	[70, 75]	2.50
[88, 92]	1.50	[65, 70]	2.75
[84, 88]	1.75	[60, 65]	3.00
[80, 84]	2.00	below 60	5.00

COURSE LEARNING EVIDENCES

To complete this course and as proof of achieving the learning outcomes, you should perform and submit the following:

Learning Evidences	Description	Course Learning Outcomes represented
Long Exams	Long Exams – 3 sets of long exams will be administered for this course for the whole semester. Topics are covered by lecture sessions. These are considered major exams that all students must take, failure to take one would incur a grade on INC.	
Lab Exams	Lab Exams – 3 sets of laboratory (lab) exams will be administered for this course for the whole semester. Topics are covered by laboratory sessions and exercises. These are considered major exams that all students must take, failure to take one would incur a grade on INC. These are also called on-the-spot programming (OTSP), where students are given a list of programming problems to solve (and code) on-the-spot.	

ABOUT THE INSTRUCTOR

In this semester, the faculty-in-charge for CMSC 18 lecture and laboratory classes is **Ms. Hanna Mae L. Limpag** (Ma'am Hannn). She is an Instructor from the Department of Mathematics, Physics, and Computer Science. Her research interest focuses on operations research and optimization.

Students can contact her through moodle, the class google chat, or the following:

- Email – hllimpag@up.edu.ph (just be mindful of your email etiquette)
- Cellular number – +639667585735 (Globe, for emergency purposes only)
- Consultation hours – every Friday 7:00 AM to 5:00PM (online or F2F). For face-to-face consultation, you can visit me at CSM Room 212.

CLASS POLICIES

Enrolling to this course means that you strongly adhere to the following class policies:

1. **Submission of Requirements and Deadlines.** Students must submit their work through the course LMS **Moodle**. Each course requirement has a deadline that the students should follow, otherwise their submission will be considered late.

Late submission deduction policies (round up to the nearest ones digit):

less than 1 day after the deadline	20% of the total score
1 to 3 days after the deadline	50% of the total score
3 to 7 days after the deadline	75% of the total score
more than 7 days after the deadline	100% of the total score (grade of zero)

2. **Exams and Quizzes.** No make-up quizzes will be given whether excused or unexcused. For missed exams, at most one make-up exam will be given for students whose absences are excused as certified by the OCS. Missing more than one exam would incur the student a score of zero (0) for the next missed exam/s. If granted, it should be taken the week after the exam was administered. Meritorious cases (bereavement, hospitalization, etc) are on a case-to-case basis. Complaints regarding the checking of exams/quizzes will only be entertained within one week of its release. The dates of examination stated above are subject to changes.
3. **Attendance.** As per university rule, a student must not absent him/herself more than 20% of the total number of classes for the term. The maximum number of allowable excused/unexcused absences is six (6) sessions. A student who exceeded these allowable absences will be FORCED to drop from the course and will merit a final grade of DRP if and only if majority of the absences are excused with proper document proof duly approved by the Office of the College Secretary; else the student will get a grade of 5.0.
It is important that students submit their excuse slip/letter (if applicable) as soon as possible after being absent. Failure to do this is a sign of irresponsibility.
4. **Dropping.** A student is officially dropped from the course if he/she/they successfully complete the dropping process. A student who stops attending classes and/or taking exams without officially dropping will be given a grade of 5.0. Note that the teacher has the right not to sign the dropping slip.
5. **Intellectual Honesty.** This course is supposed to give you knowledge and skills to better prepare you for your future career. However, this purpose will be defeated if you cheat on your work (e.g. copying of codes and submitting it as your own, etc.). Hence, all forms of academic dishonesty will not be tolerated. Any student(s) who are suspected or caught cheating will undergo investigation. If found guilty of cheating, s/he will receive a grade of 5.0 and will result in disciplinary action in accordance with university rules, policies, and procedures. Refer to Section 2 and Section 25 of the UP Diliman Faculty Manual.
6. **Technical difficulties.** Any problems (e.g. submission concerns, etc.) or other important concerns about the course should be raised to your instructor immediately. Do not hesitate to ask questions to your instructor concerning your course.

7. **Stay tuned.** Regularly check the course LMS Canvas for weekly updates (e.g. uploaded learning resources, requirements, assessments, etc.)
Further rules may be added later.

IMPORTANT INFORMATION

Academic Integrity

As a student of the University of the Philippines, you pledge to act ethically and uphold the value of honor and excellence.

You are encouraged to execute your tasks or activities truthfully and independently. You may work with others as permitted by the Instructor. With this, you understand that any suspected misconduct will be reported to the appropriate office for investigation and if established, will result in disciplinary action in accordance with University rules, policies, and procedures.

Copyright Notice

All materials in either printed or digital form are intended only for the use of the students enrolled in this course. These resources may contain works that are subject to copyright protection under RA 8293 or the “Intellectual Property Code of the Philippines”. Any reproduction and/or communication of any materials included in this course may be subject to copyright infringement and the copyright owners have the right to take legal action against such infringement.

Do not remove this notice.

Revision No		<i>Note: A revision of this course syllabus is considered when there are changes in CLOs, topics (more than 20% of the contents), and course requirements (with justification).</i>
Revised (Date)	August 1, 2024	
Revised by	Hanna Mae Limpag	Signature
Evaluated by	Assoc. Prof. Cinmayii Manliguez	Signature
Endorsed by	Assoc. Prof. Cinmayii Manliguez	Signature
Approved by	Prof. Cleto Nanola Jr.	Signature